

Sara Soldano

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PROFILE

Computational cognitive scientist with expertise in neural data analysis, EEG methodology, and human memory research. MSc candidate at the University of Edinburgh investigating rotational population dynamics in working memory using high-density neural recordings and dimensionality reduction. Combines rigorous neuroscience methods with strong programming skills to extract insight from complex behavioural and neural datasets. Interested in research roles at the intersection of brain, behaviour, and computation.

EDUCATION

The University of Edinburgh, MSc Cognitive Science *Sep 2025 – Sep 2026*

Relevant Modules: Computational Cognitive Neuroscience, Computational Cognitive Science, Machine Learning Systems, Computational Neuroscience, Accelerated NLP, Automatic Speech Recognition

Dissertation: *Examining Rotational Population Dynamics During Working Memory at a State-Conditioned Level.* Using Neuropixels primate dataset, TDR, and subspace alignment analysis to test whether rotational dynamics characterise On-state prefrontal activity at the single-trial level.

University of California, Santa Cruz, BS Cognitive Science, Minor in Computer Science *Aug 2020 – June 2024*

GPA: 3.75 • Graduated with Honors in the Major

Relevant Coursework: Cognitive Neuroscience, Computational Models, Data Structures & Algorithms, AI, Eyetracking, Design Research, Python, Computer Systems

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Samaha Lab, UC Santa Cruz *Mar 2023 – Jun 2024*

Investigated how individual alpha frequency modulates visual consciousness and temporal perception, contributing to research on the neural basis of metacognition and time perception.

- Conducted 150+ hours of EEG data collection including participant preparation, electrode placement, and real-time signal monitoring
- Designed and implemented experimental paradigms in MATLAB using PsychToolBox; independently managed full participant pipeline from recruitment through data collection
- Pre-processed and cleaned 10,000+ EEG trials, applying artifact rejection and quality control protocols to ensure dataset integrity

PROFESSIONAL EXPERIENCE

Mother Tongue, Conversational AI Model Trainer *May 2022 – May 2023*

Trained 200+ large language models across domains including tech support, history, and customer service, optimising for naturalness and accuracy

- Developed and iteratively refined natural language datasets through supervised learning feedback loops, improving model response quality
- Added structured annotations and ratings to enrich training data, contributing to more nuanced model behaviour

PUBLICATIONS & PRESENTATIONS

Publication

Morrow, A., Wilson, M., Geller-Montague, M., Soldano, S., Hajidamji, S., & Samaha, J. (2026). Individual alpha frequency predicts the sensitivity of time perception. *Imaging Neuroscience*, 4. doi:10.1162/IMAG.a.1263

Conference Abstracts

Morrow A., Wilson M., Geller-Montague M., Soldano S., Hajidamji S., Samaha J.; *The Role of Alpha Oscillations in Perceiving Visual Durations*. Journal of Vision 2024;24(10):1031. doi:10.1167/jov.24.10.1031.

Dou W., Soldano S., Volkan M., Williams M., Samaha J.; *How does subjective confidence influence multisensory integration?*. Journal of Vision 2024;24(10):477. doi:10.1167/jov.24.10.477.

SKILLS

Neural & Research Methods: EEG (setup, preprocessing, artifact rejection), PsychToolBox, eye-tracking, experimental design, participant recruitment, data pre-processing

Programming: Python, MATLAB, C, C++, JavaScript, HTML, Unix Shell, RISC-V Assembly, Frontier models, Claude Code, Cowork

Analysis & Modelling: Dimensionality reduction (TDR, PCA), computational modelling, neural population dynamics, statistical analysis, SPSS

Design: Figma, Adobe Creative Suite, DaVinci Resolve

Languages: English (native), Italian (native), Spanish (proficient)

RECOGNITIONS & CERTIFICATIONS

Dean's Honors List: Spring 2024, Summer 2023, Spring 2023, Winter 2023, Spring 2021, Spring 2020

State Seal of Biliteracy • CITI Human Subjects Research Certification • ServSafe California Food Handler